

Mikhail Razakov

Compiler Infrastructure / Language Tooling Engineer

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Compiler/language-infrastructure engineer with professional LLVM and static-analysis experience. Flagship project UniversalToolchain builds a modular compiler/runtime around typed artifact contracts, dependency/conflict resolution, deterministic pass ordering, and an immutable LanguagePlan; an architecture talk based on the project was accepted to LangDev'26.

EXPERIENCE

2026 — present

ISP RAS — Static Analysis Engineer

- Contribute to SharpChecker, ISP RAS's industrial static-analysis platform for C#/NET.

July — August
2026

MCST — Compiler Engineering Intern

- Implemented LLVM optimization work with explicit legality and unsupported-case boundaries.
- Developed loop/interprocedural analysis and LLVM IR transformations in C++23.

PROJECTS

UniversalToolchain

Modular compiler/runtime infrastructure: typed artifact contracts, dependencies/conflicts, provider selection, pass ordering, and artifact routes compile into an immutable LanguagePlan before execution.

Current exact test manifest: 1,324/1,324 passing; interpreter/CIL parity, exact package binding, and fail-closed runtime checks protect the selected execution plan.

LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effects/affine analysis and a separate legality decision; APInt models fixed-width wraparound.

29 positive/negative regression cases: global-iv, verify, structural expectations, idempotence, and observable before/after execution comparison.

x86-64 Codegen & Register Allocation Lab

A compiler backend with SSA/CFG validation, liveness/interference analysis, linear-scan and seeded simulated-annealing allocators, spills, phi lowering, and a SysV x86-64 emitter.

An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves.

SKILLS

Compiler Infrastructure

typed artifact contracts, IR/lowering pipelines, deterministic provider/pass planning, artifact routing, immutable execution plans

Compiler / IR

C++23, LLVM 22/IR, CFG/SSA, loop and interprocedural analysis, legality

Backends / Runtime

.NET, interpreter/CIL parity, SysV x86-64, register allocation, runtime composition

Verification

fail-closed checks, regression contracts, LLVM Verifier, differential execution

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.

UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.

MEPhI Junior

Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.

Moscow / remote